Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/19/2023
PI Name:	Ferl, Robert Ph.D.		
Project Title:	The Role of Ca2+ Signaling During the Early Events of Plant Adaptation to Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
PI Address 2:	Box 110690		
PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	10/20/2019	**End Date:**	10/19/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/19/2022 per NSSC information (Ed., 10/5/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gilroy, Simon Ph.D. (University of Wisconsin, Madison) Paul, Anna-Lisa Ph.D. (University of Florida, Gainesville) Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K1500		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research seeks to test the hypothesis that plants generate Ca2+ signals during transitions between different levels of gravity, signals that are trigged by the cellular gravity sensing machinery or stress response systems. Plants and cell cultures expressing the green fluorescent protein-based Ca2+ sensor GCaMP3 will be continuously monitored during a suborbital flight on Blue Origin's New Shepard spacecraft. The resultant imaging data will be used to follow Ca2+ signaling events during hypergravity-to-microgravity transitions, during sustained microgravity, and then during the return to unit gravity through the reentry profile. Plant imaging will be accomplished using the FLEX fluorescence imaging system developed by the principal investigator (PI). This equipment is flight proven for the suborbital arena. Links to the gravity and stress systems within the plant will be made through comparison of the response of wild-type plants to gravity sensing mutants (pgm1 and adg1) and to mutants in a core ROS (reactive oxygen species)/spaceflight stress response signaling element (rbohd). Comparison of measurements of callus cultures to seedlings of these lines will allow inferences to be drawn about how much observed Ca2+ signaling changes are intrinsic to cellular responses versus a requirement of action of the complex gravisensing and stress systems comprising the differentiated cells and organs of the whole plant. This work will capitalize on the combined expertise of two groups: suborbital imaging and spaceflight responses (Ferl and Paul) and Ca2+ signaling and GCaMP imaging (Gilroy and Swanson).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research seeks to test the hypothesis that plants generate Ca2+ signals during transitions between different levels of gravity, signals that are triggered by the cellular gravity sensing machinery or stress response systems. Plants and cell cultures expressing the green fluorescent protein-based Ca2+ sensor GCaMP3 will be continuously monitored during a suborbital flight on Blue Origin's New Shepard spacecraft. The resultant imaging data will be used to follow Ca2+ signaling events during hypergravity-to-microgravity transitions, during sustained microgravity, and then during the return to unit gravity through the reentry profile. Calcium mediated signal transduction is an important component of plant stress responses, and understanding how plants cope with stress in novel environments such as the transition to microgravity helps us understand stress responses in all environments, including familiar ones on Earth that affect crop health and productivity.
Task Progress:	Biology This experiment was built around the imaging of Arabidopsis thaliana plants and cell cultures expressing a green fluorescent protein-based biosensor (GCaMP3) that can detect Ca2+ signals within the plant. Therefore, initial work focused on characterizing the mutant lines to be used and then generating the stable transgenic lines from these plants. This experiment was built around the imaging of Arabidopsis thaliana plants and cell cultures expressing a green fluorescent protein-based biosensor (GCaMP3) that can detect Ca2+ signals within the plant. Therefore, initial work focused on characterizing the mutant lines to be used and then generating the stable transgenic lines from these plants. We have tested the mechanically-induced Ca2+ response in the GCaMP reporter lines to be flown. Mechanical forces are likely an important component of the stresses these plants will experience during flight. In these ground-based control experiments, both wild type and the RBOHD mutants showed robust touch-induced Ca2+ increases with near identical kinetics, suggesting that any differences seen in Ca2+ dynamics during flight will be unlikely to be the effect of differences in their mechanoperception systems. Experimental Flight This payload was contracted with Blue Origin, a flight provider, through the University of Florida. This experiment was hosted in the European Space Agency FLuorescence EXplorer (FLEX) imager, which has been flight proven on four previous sub-orbital flights, including three flights on Blue Origin. This hardware hosts two green fluorescent protein (GFP) sensitive cameras, including a new thermoelectric (TE) cooled camera for stabilized image exposure. This camera also expands the imaging area to allow for a plate and a half imaging area, which will increase science return from the payload. FLEX was also fitted with an MSR Electronics data logger for environmental flight conditions and a Surface 8 tablet to record data in real time during the flight. Biology Flown For this flight experiment, the payload hosted three square petri plates for imaging and four square petri plates for field harvest post-flight, followed by biochemical analysis during processing. The imaging plates are comprised of lines developed for this experiment by the Gilroy lab (biology section of this report) and previously flown lines from the Ferl-Paul lab. The passively flown plates that were harvested for analysis, that have been flown before and were the second replicate for the transcriptomic experiment. Prior to flight, imaging plates were opened, and the media as well as plants were allowed to be exposed to air for approximately 20 hours prior to flight. This facilitated imaging plants without the lid on the plate and allows plants recovery time when exposed to local atmospheric humidity. Flight Operations Due to requirements of the flight provider, Blue Origin, hardware must be ready and turned over approximately 6 hours prior to launch. At T-6 hours, plates were mounted in FLEX and secured with Velcro. The payload was then handed off to Blue Origin for loading and flight. Images were collected every second with a 995000ms exposure on both cameras. Imagers were expected to be turned on 7 minutes prior to launch. However, due to holds during flight preparation we were able to capture approximately 45 minutes of imaging prior to launch. Environmental Data An MSR145 data logger was mounted inside the FLEX payload locker to record environmental data from handover to launch. The MSR recorded Relative Humidity (RH) and Temperature (T) every 5 seconds from 1:30am local time until the payload was recovered (approximately 12:30pm local time.). Temperatures were recorded as expected, held constant overnight with increases during daylight expected as the payload cabin is not temperature controlled. In addition to temperature and relative humidity, the MSR recorded 3-axis acceleration data at a rate of 25 hz. Image Data The FLIR Grasshopper/Katydid camera collected a total of 3505 frames covering pre-flight, flight, and landing. The Imperx Cheetah camera collected a total of 3519 frames covering pre-flight, flight, and landing. Results

	Data analysis for the flight is ongoing but patterns of GFP expression have emerged. Image data processing has shown limited consistent results, indicating a specific response to spaceflight by plants.
Bibliography Type:	Description: (Last Updated: 11/27/2022)